

CODE

```
#include<iostream>

using namespace std;

struct node
{
    int data;
    node *pre;
    node *next;
};

class A
{
public:
    node *start,*end,*temp,*nextaftercur,*cur;
    A()
    {
        start=NULL;
        end=NULL;
    }
    void insertion(int value,int pos)
    {
        pos--;
        if(pos==0)
        {
            if(start==NULL)
            {
                start=new node;
                start->data=value;
                start->pre=NULL;
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        start->next=NULL;
    }
    else
    {
        temp=new node;
        temp->data=value;
        temp->pre=NULL;
        temp->next=start;
        start->pre=temp;
        start=temp;
    }
}
else
{
    int i=0;
    cur=start;
    while(cur->next!=NULL)
    {
        if(i==pos-1)
        {
            nextaftercur=cur;
            nextaftercur=nextaftercur->next;
            temp=new node;
            temp->data=value;
            temp->next=cur->next;
            cur->next=temp;
            temp->pre=cur;
            nextaftercur->pre=temp;
            return;

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        }
        cur=cur->next;
        i++;
    }
    if(cur->next==NULL&& i==pos-1)
    {
        temp=new node;
        temp->data=value;
        temp->next=NULL;
        temp->pre=cur;
        cur->next=temp;
        end=temp;
    }
    else
    {
        cout<<"Position Is Out Of Bound For Insertion"<<endl;
        return;
    }
}

void del(int pos)
{
    pos--;
    temp=start;
    cur=start;
    if(pos==0)
    {
        cur=cur->next;
        cur->pre=NULL;
    }
}

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        start=cur;
        delete temp;
    }
else
{
    int i=0;
    while(cur->next!=NULL)
    {
        if(i==pos)
        {
            nextaftercur=cur;
            nextaftercur=nextaftercur->next;
            nextaftercur->pre=temp;
            temp->next=cur->next;
            delete cur;
            return;
        }
        temp=cur;
        cur=cur->next;
        i++;
    }
    if(cur->next==NULL&& i==pos)
    {
        temp->next=NULL;
        delete cur;
        return;
    }
else
{

```

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        cout<<"Position Is Out Of Bound For Deletion"<<endl;
    }
}

void display()
{
    cur=start;
    while(cur->next!=NULL)
    {
        cout<<cur->data<<" ";
        cur=cur->next;
    }
    cout<<cur->data<<endl;
}

};

int main()
{
    A a;
    a.insertion(4,1);
    a.display();
    a.insertion(3,1);
    a.display();
    a.insertion(2,3);
    a.display();
    a.insertion(1,7);
    a.display();
    a.insertion(1,3);
    a.display();
    cout<<endl<<"//Deletion"<<endl<<endl;
}

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a.del(3);

a.display();

a.del(3);

a.display();

a.del(1);

a.display();

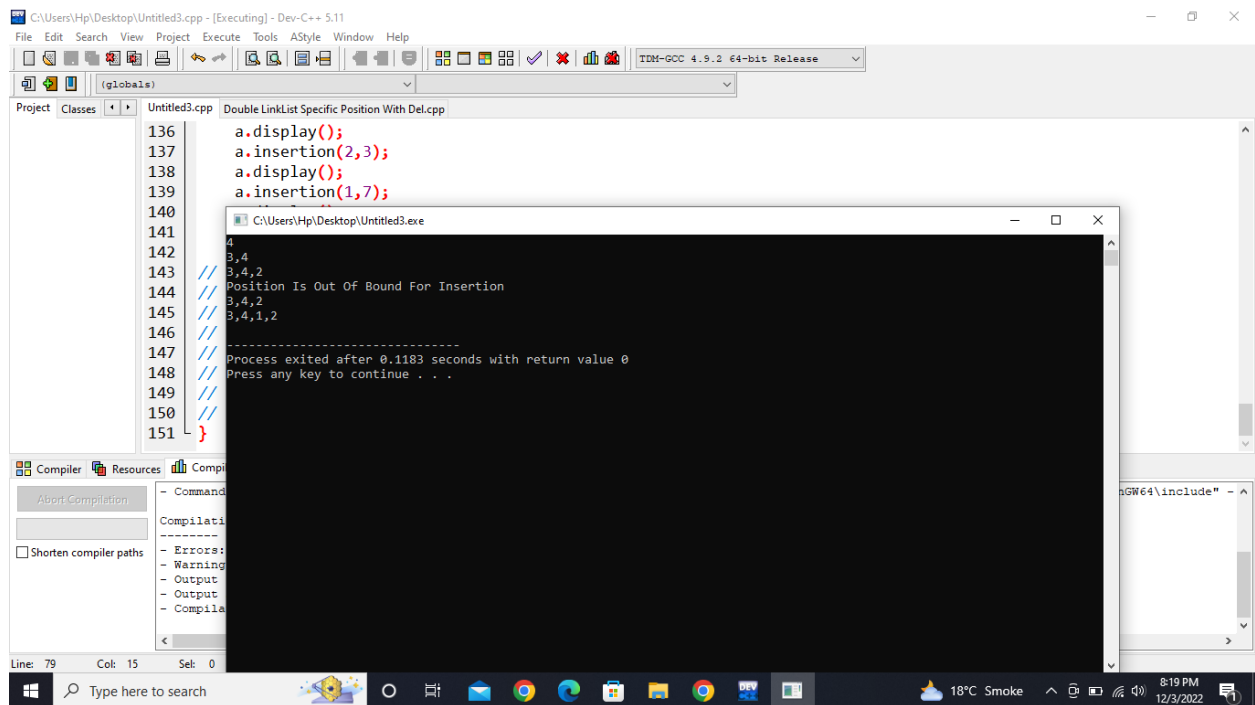
a.del(2);

a.display();

}

```

For Insertion



For Deletion

